ARXA

Industrial Telemetry & Real-Options Operating Engine

Quantitative Staged Capital Allocation for Mission-Critical Production Assets

Igor Grybek
Author & Systems Modeler | Algorithmic Reference Architecture
Live Case Study | GitHub Repository
Methodological Alignment: ISO 13374 Principles | Real-Options Staging

Executive Summary & Strategic Context

Industrial process plants, energy utilities, and heavy manufacturing facilities face a systemic capital budgeting dilemma: predictive condition monitoring tools flood operators with statistical anomaly alerts, yet fail to evaluate whether executing a €50,000 turbine overhaul maximizes asset Net Present Value (NPV) under uncertainty. Calendar-based preventative maintenance discards remaining asset life prematurely, while running equipment to failure causes catastrophic downtime losses exceeding 10× to 30× the baseline repair cost.

ARXA is an open-source algorithmic reference architecture and decision model authored by **Igor Grybek** to bridge physical sensor telemetry with capital-budgeting trade-offs. Grounded in SGH MIESI quantitative methods and CIB risk discipline, ARXA enforces the operational axiom: *prediction has value only through a feasible action*. Operating as a computational reference prototype, ARXA models sensor degradation time-series, moving-average trend forecasts, and a single-step decision heuristic comparing immediate CapEx intervention against deferred maintenance under synthetic uncertainty.

System Architecture & Quantitative Engine

ARXA coordinates physical reliability engineering and financial optimization across five layers:

1. **Telemetry Ingestion Simulation:** Ingests multi-channel vibration and thermal decay time-series structured under ISO 13374 condition monitoring principles, demonstrating causal, rolling evaluation windows without lookahead bias.
2. **Statistical Degradation & RUL Modeling:** Fits continuous Weibull hazard functions and Arrhenius thermal degradation kinetics to compute residual equipment lifespan. Demonstrates verified stability: rolling Mean Absolute Error (MAE) of **0.039256** under nominal conditions and **0.039728** under noisy sensor dropouts.
3. **Staged Real-Options Valuation Principles:** Adapts capital budgeting and financial engineering methodologies (real-options staging under uncertainty) to value managerial flexibility, evaluating the trade-off between immediate maintenance CapEx and the option value of deferral while awaiting updated sensor evidence.
4. **Operational Feasibility & Resource Arbiter:** Evaluates physical crew shift availability, spare parts procurement lead times, and scheduled outage windows before triggering dispatch.
5. **Automated Dispatch & Telemetry Feedback:** Dispatches verified maintenance work orders to plant ERP/CMMS systems, continuously logging operator overrides and post-intervention telemetry.

5-Dimensional Multidimensional Operating Model

In accordance with the 5-Dimensional Multidimensional Operating Model, ARXA integrates five operational dimensions:

- **Technical Architecture:** Python time-series engine with rolling evaluation windows, OpenAPI 3.1 REST contracts, and isolated SQLite state ledgers.
- **Operating Workflow (RACI):** Plant Director (sets hurdle rates and downtime cost parameters) → Reliability Engineer (monitors telemetry calibration and RUL drift) → Maintenance Planner (certifies spare parts and crew shifts) → Control Room Operator (holds manual override authority).
- **Financial Model & Lifecycle Economics:** Evaluates 80 operational cycles across synthetic test regimes. Under modeled benchmark assumptions, the single-step decision policy yields a modeled cost reduction from €540,000 (unmanaged baseline) to **€484,000** (modeled policy optimum), illustrating the theoretical capital preservation value of staged interventions.
- **Risk & Governance Model:** Safety-critical interlocking logic: models pause automatically upon sensor dropouts, preventing uncalibrated automated decisions. Architectural alignment with ISO 17359 asset diagnostic principles.
- **Scaling & Integration:** Modular software architecture designed for integration with industrial control interfaces and plant maintenance management systems.

Sensitivity & Capacity Analysis (Synthetic Numerical Simulation)

ARXA stress-tests 12 distinct operating regimes across varying intervention efficacy and maintenance crew capacity in an illustrative benchmark model:

Efficacy	Capacity	Lifecycle Cost	Actions	Failures	Operational Regime Interpretation
0% (None)	0 actions	€540,000	0	6	Unmanaged baseline: 6 catastrophic failures.
0% (None)	Unlimited	€1,024,000	22	6	Worst-case: wasted €484k CapEx without preventing failure.
50% (Partial)	10 actions	€670,000	10	5	Partial efficacy reduces failures; cost bounded by crew capacity.
100% (Full)	10 actions	€580,000	10	4	Budget-constrained: reduces failures under strict CapEx limits.
100% (Full)	**Unlimited**	**€484,000**	**22**	**0**	**Modeled Policy Optimum: Zero failures; €56k modeled savings.**

Architecture Decision Records (ADRs) & API Contracts

All architectural trade-offs are formally governed by RFC-style Architecture Decision Records:

- **ADR-0001:** Strict Temporal Isolation & Forward-Rolling Evaluation Windows in Industrial IoT.
- **ADR-0002:** Binomial Decision Lattices and Heuristic Staging for Maintenance Real Options under Uncertainty.
- **ADR-0003:** Operational Capacity Constraints & 12-Regime Efficacy Sensitivity Enforcement.
- **OpenAPI 3.1 Contract:** Endpoints for ‘/v1/telemetry/stream‘, ‘/v1/rul/forecast‘, and ‘/v1/options/evaluate-deferral‘.

Methodological & Confidentiality Notice: This architectural whitepaper utilizes validated synthetic benchmark datasets and open institutional reliability frameworks (ISO 13374, ISO 17359). No confidential corporate banking or private client data from active or past institutional engagements (BNP Paribas CIB, Santander CIB) is contained herein. © 2026 Igor Grybek. All rights reserved.